(5) (i) $(AB)^{-1} = B^{-1}A^{-1}$. (ii) $(A^{-1})^{-1} = A$

(6) If A is a non-singular square matrix of order n, then

(i) $\left(\text{adj } A\right)^{-1} = \text{adj}\left(A^{-1}\right) = \dfrac{1}{|A|}A$ (ii) $|\text{adj } A| = |A|^{n-1}$

(iii) $\text{adj}\left(\text{adj } A\right) = |A|^{n-2}A$ (iv) $\text{adj}(\lambda A) = \lambda^{n-1}\text{adj}(A)$, λ is a nonzero scalar

(v) $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$ (vi) $(\text{adj } A)^T = \text{adj}\left(A^T\right)$

(vii) $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

(7) (i) $A^{-1} = \pm\dfrac{1}{\sqrt{|\text{adj } A|}}\text{adj } A$. (ii) $A = \pm\dfrac{1}{\sqrt{|\text{adj } A|}}\text{adj}\left(\text{adj } A\right)$.

(8) (i) A matrix A is orthogonal if $AA^T = A^TA = I$

(ii) A matrix A is orthogonal if and only if A is non-singular and $A^{-1} = A^T$

(9) Methods to solve the system of linear equations $AX = B$

(i) By matrix inversion method $X = A^{-1}B, |A| \neq 0$

(ii) By Cramer's rule $x = \dfrac{\Delta_1}{\Delta}, y = \dfrac{\Delta_2}{\Delta}, z = \dfrac{\Delta_3}{\Delta}, \Delta \neq 0$.

(iii) By Gaussian elimination method

(10) (i) If $\rho(A) = \rho([A\,|\,B])$ = number of unknowns, then the system has unique solution.

(ii) If $\rho(A) = \rho([A\,|\,B]) <$ number of unknowns, then the system has infinitely many solutions.

(iii) If $\rho(A) \neq \rho([A\,|\,B])$ then the system is inconsistent and has no solution.

(11) The homogenous system of linear equations $AX = 0$

(i) has the trivial solution, if $|A| \neq 0$.

(ii) has a non trivial solution, if $|A| = 0$.

ICT CORNER

https://ggbm.at/vchq92pg or Scan the QR Code

Open the Browser, type the URL Link given below (or) Scan the QR code. GeoGebra work book named "12th Standard Mathematics" will open. In the left side of the work book there are many chapters related to your text book. Click on the chapter named "Applications of Matrices and Determinants". You can see several work sheets related to the chapter. Select the work sheet "Application Matrices−Problem"

S62YB

12th_Maths_EM_Vol1_CH 1_Matrices.indd 51 16-12-2022 12:21:24